

Car Insurance and Pricing Estimation System

CENG 476

Ata Doruk Çakmak

202111030

Utku Bora Efil

202211023

Öykü Çoban

202111066

Date:

Abstract

This project presents a machine learning-based system for estimating car insurance costs and market prices. Users can input vehicle details—such as year, mileage, fuel type, and transmission—to receive on-demand predictions using trained Random Forest models. Developed with Python and Streamlit, the application incorporates data preprocessing, encoding of categorical variables, and interactive visualizations to support intuitive exploration and analysis. Although the insurance pricing logic is based on a realistic custom formula rather than real-world data, the system serves as a valuable tool for individuals and educational purposes. It demonstrates how machine learning can aid data-driven decision-making in vehicle valuation and insurance estimation.

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1. Introduction

Estimating car insurance costs and vehicle market prices accurately is an important concern for car owners, buyers, and resellers. Traditional methods of evaluation are often manual, inconsistent, and lack scalability. To address this, machine learning offers a data-driven approach that can generate more consistent and automated predictions.

This project introduces a web-based application that allows users to enter vehicle information and receive instant predictions for both insurance premiums and market prices. Built with Python and Streamlit, the system uses a trained Random Forest algorithm to make predictions based on key features such as year, mileage, fuel type, and transmission. Additionally, the application includes interactive data visualizations that help users explore how different factors affect pricing and risk.

While the insurance pricing logic is modeled using a custom risk factor formula rather than real-world insurance datasets, the system demonstrates the potential of combining machine learning with user-friendly interfaces to support vehicle evaluation and pricing education.

2. Technologies Used

Programming Language: Python

Web Framework: Streamlit

Data Handling and Visualization Libraries: pandas, NumPy (data processing) matplotlib, seaborn (data visualization)

Machine Learning: scikit-learn

Model used: RandomForestRegressor for both car price prediction and insurance estimation

Model Persistence: joblib (used to save and load trained models)

3. Project Structure

data_processor.py: handles preprocessing and feature engineering

model_trainer.py: trains and saves the Random Forest model

utils.py: includes helper functions for risk calculation, formatting, and validation

app.py: Streamlit interface connecting all components

4. Data Processing

The dataset used in this project contains multiple features related to vehicles, such as car make, manufacturing year, mileage, fuel type, engine size, and ownership history. The data processing pipeline includes several crucial steps:

Handling missing or inconsistent values to ensure data quality.

Encoding categorical variables using label encoders to convert text categories into numerical values suitable for machine learning models.

Feature scaling where applicable to normalize numerical data ranges.

Splitting the dataset into training and testing subsets for model validation.

Basic exploratory data analysis (EDA) was performed to understand distributions, detect outliers, and visualize correlations among features.

These preprocessing steps were essential to improve the models' learning efficiency and accuracy.

5. Model Training

Machine learning models were trained to predict two primary targets: car insurance premiums and vehicle market prices. The training process involved:

Using the preprocessed dataset to train regression models capable of continuous value prediction.

Splitting the data into training and testing sets to evaluate model performance.

Applying algorithms such as Random Forest Regression (specific model details will be elaborated later).

Evaluating models using metrics like Root Mean Squared Error (RMSE) and R-squared (R^2) scores.

Saving the trained models to the models/ directory for subsequent use in real-time prediction within the web application.

The models serve as the predictive backend, loaded dynamically by the application during user interaction.

6. Application Interface

The user interface is developed using Streamlit, which provides a simple and interactive web platform. Key features include:

Input widgets such as dropdown menus and sliders that allow users to specify car details (e.g., year, brand, fuel type, mileage).

Upon input submission, the app uses the trained models to generate and display predicted insurance costs and market prices instantly.

Visualizations including charts and graphs enable users to explore trends and relationships within the dataset, enhancing understanding and trust in the predictions.

The interface is designed to be intuitive and accessible for both individual car owners and insurance company representatives.

7. Challenges and Limitations

This project faced several constraints during development:

Due to environmental and resource limitations, full execution of model training and deployment was not possible on the developer's local system.

The application relies on pre-trained models; therefore, real-time retraining or updates were not implemented.

Some external dependencies and dataset files were unavailable, restricting end-to-end testing.

Despite these limitations, the codebase and architecture were structured to support full functionality once the required resources are provided.

8. Conclusion

This project lays a foundational framework for an automated car insurance and pricing prediction system powered by machine learning. Although the system's execution was partial, the modular design and prepared scripts demonstrate strong potential for practical deployment. Future work could include:

Expanding the dataset and retraining models for improved accuracy.

Integrating real-time data sources or APIs.

Enhancing the web interface with more user customization options.

Implementing continuous model updates for dynamic market conditions.

With continued development, this system could become a valuable tool for users seeking data-driven decisions in vehicle valuation and insurance planning.